RETAIL SALES ANALYSIS

Excel, SQL, Python & Power BI

Abstract

This project focuses on building an end-to-end Retail Sales Analytics solution using Excel, SQL, Python, and Power BI to transform raw retail transaction data into meaningful business insights. The objective was to analyze sales performance, customer behavior, product trends, and inventory status while creating an interactive business intelligence dashboard.

In the **Excel** phase, data preprocessing and cleaning activities were performed, including handling missing values, removing duplicate records, standardizing column names, validating data formats, creating calculated fields, and preparing structured CSV files for database integration. Pivot tables and lookup functions were utilized to generate preliminary sales summaries and ensure data quality.

In the **SQL** phase, a relational database was designed and populated using the cleaned datasets. Advanced SQL queries involving JOINs, Aggregate Functions, Subqueries, CASE statements, and GROUP BY operations were executed to analyze store performance, customer purchases, product sales, staff productivity, and inventory levels. The database served as the foundation for further analytical processing.

Using **Python**, data was extracted from SQL databases and analyzed through Exploratory Data Analysis (EDA). Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to examine sales patterns, customer behavior, and business trends through statistical summaries and visualizations. Customer segmentation was performed using RFM (Recency, Frequency, Monetary) analysis to classify customers based on purchasing behavior and generate actionable insights.

Finally, **Power BI** was used to create an interactive dashboard by connecting directly to the SQL database. Various visualizations, including sales trend analysis, top-performing products, customer purchase patterns, store-wise sales distribution, staff performance metrics, inventory monitoring, and customer segment analysis, were developed. Interactive slicers, KPIs, and drill-down capabilities enabled dynamic exploration of business performance.

The project successfully demonstrates the complete data analytics lifecycle, from data cleaning and database management to advanced analytics and business intelligence reporting, enabling data-driven decision-making in the retail domain.